CODE ALPHA

TASK 2 REPORT

Structural Simulation (FEA) Report

Task 2 — Finite Element Analysis of an L-Bracket

Linear-elastic, 2D plane-stress analysis · solver verified against beam theory

1. Objective

Perform a finite element analysis (FEA) on a simple structural component — an L-shaped mounting bracket — to evaluate its response under different loads and boundary conditions. The study reports the stress distribution (von Mises), the deformation (displacement field), and the factor of safety against yielding for three distinct load/restraint scenarios, and draws design observations from the results.

2. Component, Material and Mesh

The component is an aluminium L-bracket modelled in 2D plane stress (constant out-of-plane thickness). A fillet is included at the re-entrant inner corner, which is the expected critical region. Properties and discretisation are summarised below.

Property	Value
Component	L-bracket — vertical leg 25 × 100 mm, horizontal leg 80 × 25 mm
Inner corner fillet	R 8 mm
Out-of-plane thickness	10 mm
Material	Aluminium 6061-T6
Young's modulus, E	68.9 GPa
Poisson's ratio, v	0.33
Yield strength, Sy	276 MPa
Analysis type	Linear elastic, static, 2D plane stress
Element type	3-node constant-strain triangle (CST)
Mesh size	1,586 nodes / 2,950 elements

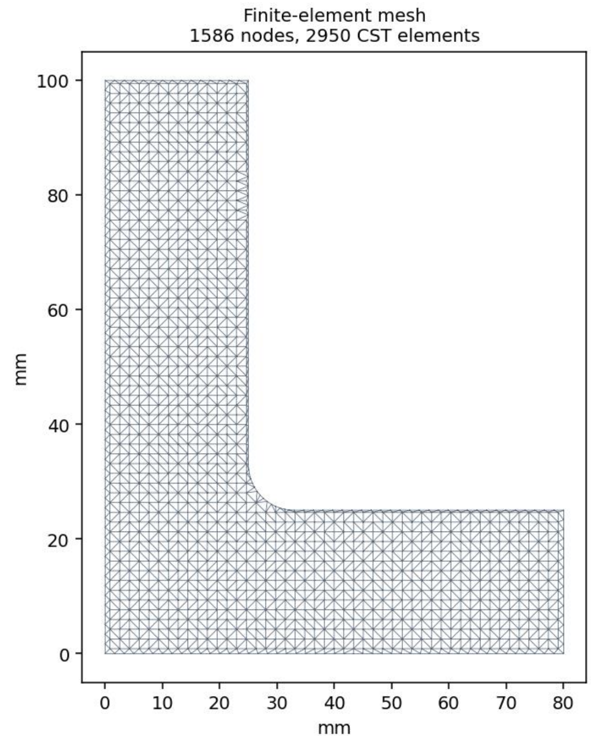

Figure 1 — Finite-element mesh of the L-bracket (refined enough to resolve the fillet).

3. Loads and Boundary Conditions

Three cases were analysed to study how the bracket responds to different loading directions and different restraints. In every case a 600 N load is applied to the end face of the horizontal arm; the difference lies in the load direction and in which surface is held fixed (clamped, all displacements = 0).

- Case 1 : Vertical (downward) 600 N tip load; the top mounting flange is fixed (as if bolted to a wall).
- Case 2 : Horizontal (pull-out) 600 N tip load; the same top flange is fixed. This changes the load direction only.
- Case 3 : Vertical 600 N tip load, but now the entire back face of the vertical leg is fixed. This changes the boundary condition only.

L-bracket — geometry, loads & boundary conditions (6061-T6 Al, t = 10 mm, inner fillet R8)

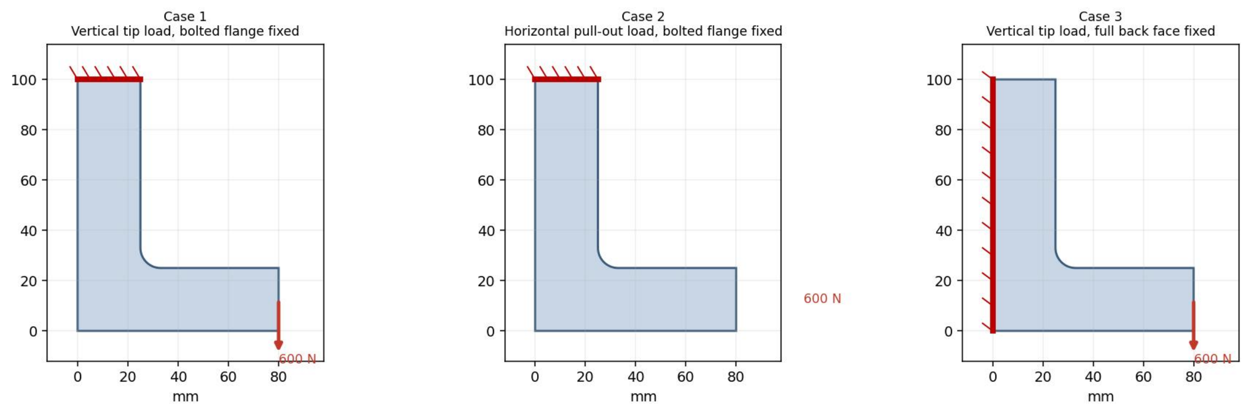

Figure 2 — Geometry, applied loads (red arrows) and fixed supports (hatched) for the three cases.

4. Solver Verification

Before trusting the bracket results, the solver was verified on a problem with a known closed-form answer: a slender cantilever beam (100 × 10 × 10 mm) with a 100 N transverse tip load. The FEA tip deflection and peak bending stress were compared with Euler–Bernoulli beam theory.

Quantity	FEA	Theory	Difference
Tip deflection	0.559 mm	0.581 mm	−3.7%
Peak bending stress	58.1 MPa	60.0 MPa	−3.2%

Agreement within ~4% confirms the formulation, units and boundary handling are correct. The small under-prediction of deflection is the expected mild over-stiffness of linear CST elements.

5. Results

For each case the figure below shows, left to right: von Mises stress on the deformed shape (with the undeformed outline ghosted), the displacement magnitude, and the factor-of-safety map (green = safe, red = at yield).

Case 1 — Vertical load, flange fixed

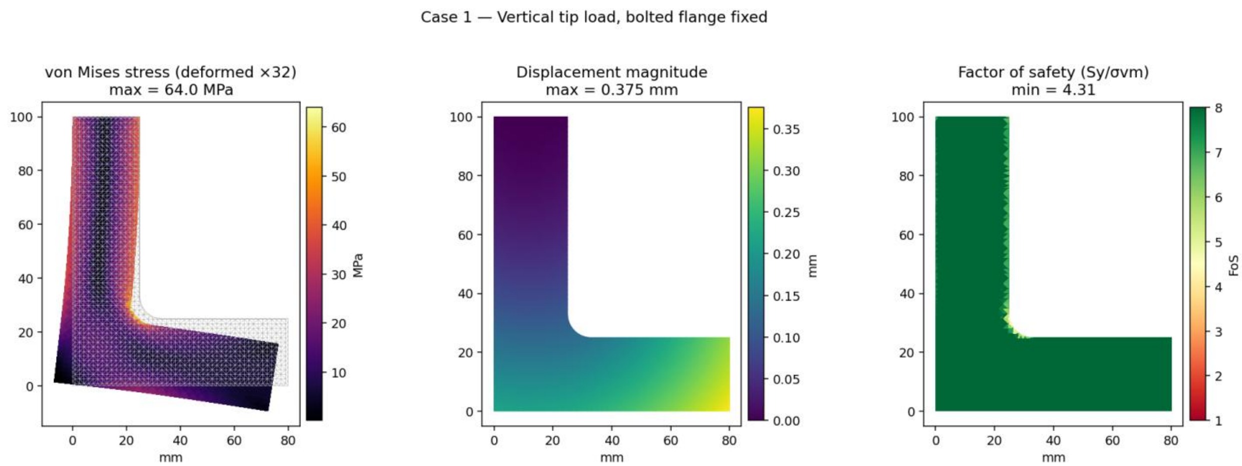

Figure 3 — Case 1 results. Max von Mises 64.0 MPa at the fillet; max deflection 0.376 mm; min FoS 4.31.

The bracket bends like a cantilever. Stress peaks sharply at the inner fillet, and the largest displacement is at the loaded tip. This is the most severe of the three cases.

Case 2 — Horizontal load, flange fixed

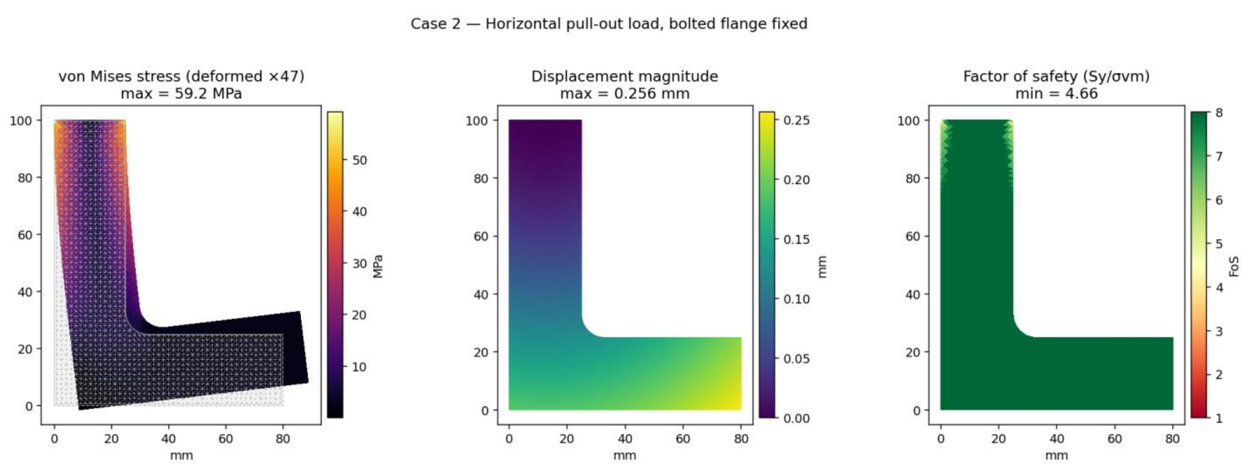

Figure 4 — Case 2 results. Max von Mises 59.2 MPa; max deflection 0.256 mm; min FoS 4.66.

Pulling the arm outward loads the vertical leg in combined tension and bending. Peak stress is again at the fillet but slightly lower, and the bracket is stiffer in this direction than under the vertical load.

Case 3 — Vertical load, back face fixed

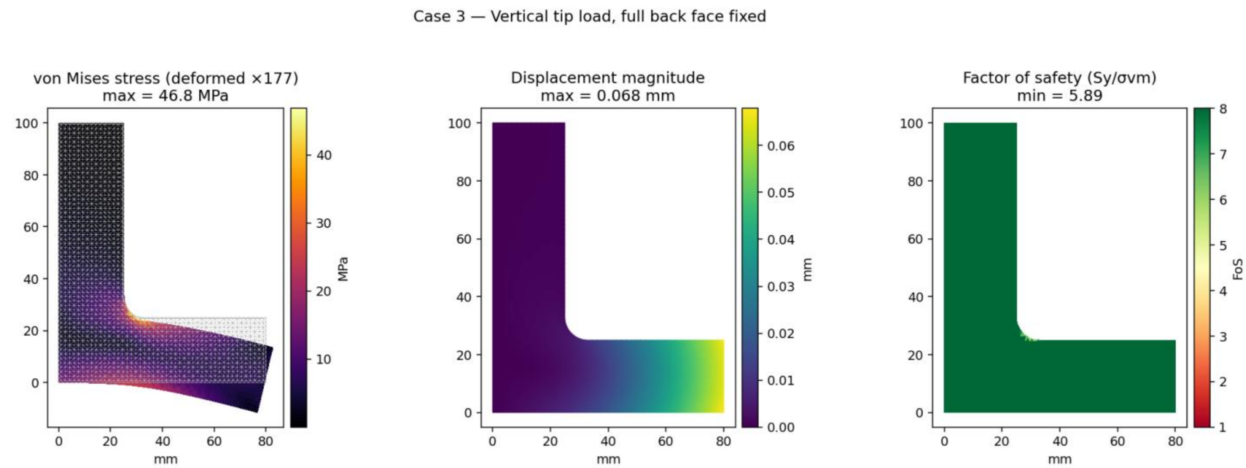

Figure 5 — Case 3 results. Max von Mises 46.8 MPa; max deflection 0.068 mm; min FoS 5.89.

Clamping the whole back face instead of just the flange removes most of the vertical leg's flexibility, so only the horizontal arm bends. Deflection drops dramatically and peak stress falls, even though the applied load is identical to Case 1.

Summary of all cases

Case	Loading & restraint	Max σ_vm (MPa)	Max defl. (mm)	Min FoS
1	Vertical 600 N, flange fixed	64.0	0.376	4.31
2	Horizontal 600 N, flange fixed	59.2	0.256	4.66
3	Vertical 600 N, back face fixed	46.8	0.068	5.89

6. Observations and Discussion

- The inner fillet is the critical location in every case — the re-entrant corner concentrates stress. A sharp (un-filleted) corner would read substantially higher, so the fillet is doing real work; increasing its radius would lower the peak stress further.
- Load direction matters: the vertical tip load (Case 1, 64.0 MPa) is more damaging than the equal horizontal load (Case 2, 59.2 MPa) because it bends the long, slender vertical leg most effectively.
- Boundary conditions dominate stiffness: simply changing the restraint from the top flange (Case 1) to the full back face (Case 3) cut deflection by ~5.5× (0.376 → 0.068 mm) and reduced peak stress by ~27%, for the very same load. How a part is mounted can matter as much as how it is loaded.
- All three cases keep the factor of safety above 4 against yield, so the bracket is safe for a 600 N static load with comfortable margin. The governing (worst) case is Case 1 with FoS = 4.31.
- The results are trustworthy in trend and magnitude: the same solver reproduced beam theory to within ~4%.

7. Limitations and Recommendations

- Analysis is linear-elastic, static and 2D plane stress; it does not capture out-of-plane effects, contact at the bolts, residual stress or large deflection.
- Linear CST elements are slightly over-stiff, so true deflections are marginally higher and the fillet peak stress would rise somewhat with mesh refinement or quadratic elements. A local mesh-refinement convergence check at the fillet is recommended.
- If the load is cyclic, a fatigue assessment at the fillet should follow, since that is where cracks would initiate.
- For a final design, confirm with a 3D model and model the actual bolted flange (bolt holes / pre-tension) rather than an idealised fixed edge.

8. Conclusion

The L-bracket was analysed under three combinations of load direction and restraint. Stress consistently concentrates at the inner fillet, the vertical load is the most demanding, and the restraint condition strongly governs both stiffness and stress. With a minimum factor of safety of 4.31, the bracket safely carries the 600 N static load, and the FEA workflow itself was verified against an analytical benchmark.